

Comprehensive Technical Architecture, API Ecosystem, Algorithms & Pitch Defense Dossier

This dossier provides an exhaustive technical and algorithmic breakdown of the PlanZo platform. Prepared for engineering project viva examinations, hackathon pitching, and technical evaluations, this document details all system languages, full-stack framework choices, server-side APIs, mathematical formulations, attendance algorithms, and structured cross-examination defenses.

KEY PROJECT SPECIFICATION SUMMARY

Tech Stack: TypeScript 5+, React 19, Vite 8, Tailwind CSS v4, Express.js 4, Google GenAI SDK (Gemini 3.8 Flash)
Architecture: Client SPA with Server-Side AI API Proxy, LocalStorage Resilience & Heuristic Fallbacks

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PROJECT DETAILS & REVISION RECORD

System: PlanZo Autonomous Student OS | Version: 2.4.0 (Production Build)

Target Audience: B.Tech Engineering Undergraduates | License: Open Academic Software

1. System Architecture & Programming Languages

PlanZo is architected as an industrial-grade full-stack TypeScript application. Every layer - from client DOM manipulation to server-side AI prompt orchestration - shares standardized TypeScript schemas, eliminating type impedance mismatches.

Frontend Layer

- TypeScript (Strict Mode): Enforces compile-time contracts across 100% of data models (AttendanceItem, ScheduledTask, SyllabusUnit, ExamMilestone). Prevents null/undefined crashes during critical student scheduling and attendance simulations.
- React 19 (Component Hierarchy): Leverages modern functional components, standard hooks (useState, useEffect, useMemo, useCallback), and centralized Context API (AppContext). Decouples visual representation from business and mathematical logic.
- Vite 8 & Tailwind CSS v4: Provides sub-millisecond hot module reloading and optimized production tree-shaking. Tailwind v4 delivers a high-contrast B.Tech productivity aesthetic (Black, White, Neutral Gray, Teal accents) with zero CSS bloat.
- Motion & Audio Synthesis: Uses lightweight physics animations (motion) for smooth layout transitions and native HTML5 Web Audio API synthesizers for instant task check-off feedback without external audio latency.

Backend Layer

- Node.js & Express.js (v4.21): Lightweight, high-throughput REST API runtime. Houses application proxy routes (/api/chat, /api/recalibrate, /api/syllabus-planner), isolating API tokens from client exposure.
- Google GenAI TypeScript SDK (@google/genai v2.4.0): Direct server integration with Google Gemini 3.8 Flash. Executes prompt-engineered requests with structured JSON schema constraints, telemetry tracking headers, and instant fallback pipelines.
- TSX Development Engine: Native TypeScript execution engine executing server.ts directly with zero manual compilation step, mirroring modern Node 22 ES module standards.

Technology Comparison & Stack Summary

Tier / Module	Technology	Version	Core Architectural Purpose
Client Framework	React & DOM	19.0.1	Declarative UI, concurrent rendering, virtual DOM
Type System	TypeScript	7.0.2	Strict compile-time verification across client & server
Styling System	Tailwind CSS	4.3.3	Utility-first tokens, dark/light variants, responsive grid
Build Bundler	Vite	8.3.0	Rollup bundling, asset compilation, ES module dev server
Server Runtime	Node.js + TSX	22.x / 4.21	Express REST proxy, process management, API endpoints
AI Model Engine	Gemini 3.8 Flash	GenAI 2.4.0	Conversational campus mentorship & schedule recalibration
Persistence	LocalStorage API	Native HTML5	Client-side zero-latency session and data resilience

2. APIs & Request-Response Communication

PlanZo implements a strict client-server separation. The browser client never interacts directly with external AI services, protecting user credentials, ensuring rate-limiting compliance, and maintaining predictable JSON contracts.

Complete API Endpoint Catalog

- POST `/api/chat`: Powers the Sarthi AI Senior Campus Mentor. Receives conversation message history alongside the student profile, current semester, college, timetable density, and habit streaks. System instructions ground the AI in realistic campus jargon (75% criteria, PYQs, vivas, internal marks) to deliver actionable 80/20 study plans.
- POST `/api/recalibrate`: Powers the Guilt-Free Schedule Recalibration engine. When a student marks a task as missed, the endpoint receives remaining daily tasks and cognitive weights, returns a reassuring 1-sentence summary, inserts a restorative buffer, and shifts intensive tasks.
- POST `/api/syllabus-planner`: Generates high-yield study chunks based on days remaining before examinations and syllabus weightages.

Detailed API Request-Response Lifecycle

Step 1: User Action Trigger - The student clicks "Start Recalibration" or asks a question in Sarthi AI.

Step 2: Context Aggregation - AppContext serializes current timetable, attendance percentages, and profile into a lightweight payload.

Step 3: Client Fetch Call - A POST request is dispatched to `/api/*` with Content-Type: `application/json`.

Step 4: Express Server Middleware - Express parses the body, injects student context, and validates schema.

Step 5: Google GenAI Invocation - The server calls `ai.models.generateContent` with model: "gemini-3.8-flash".

Step 6: Fallback Safeguard - If network is offline or quota is exceeded, an intelligent local heuristic algorithm provides instant responses.

Step 7: State Update & Storage - Client receives JSON, fires audio/confetti feedback, and persists state in LocalStorage.

SECURITY & CREDENTIAL ISOLATION GUARANTEE

PlanZo strictly forbids embedding `GEMINI_API_KEY` in client bundles (Vite `VITE_*` variables). All external AI calls pass through the Express reverse-proxy server,

3. Algorithmic Formulations & Mathematical Models

At the heart of PlanZo are deterministic mathematical algorithms engineered specifically for Indian university engineering guidelines (specifically AICTE and autonomous universities enforcing mandatory 75% attendance).

Algorithm 1: Academic Attendance Debarment & Buffer Calculus

Let A = number of attended classes in a subject, and T = total classes conducted so far. The institution mandates attendance fraction $P_{\min} = 0.75$.

SAFE BUNK MARGIN FORMULA (WHEN CURRENT ATTENDANCE $\geq 75\%$)

$$\text{Safe Bunks} = \text{floor}((4 * A - 3 * T) / 3)$$

Calculates the exact integer number of future consecutive classes a student can safely miss without dropping below 75%.

RECOVERY ATTENDANCE REQUIREMENT (WHEN CURRENT ATTENDANCE $< 75\%$)

$$\text{Required Lectures} = \text{ceil}(3 * T - 4 * A)$$

Calculates the exact integer number of consecutive future lectures the student must attend to restore their attendance to 75%.

Algorithm 2: Guilt-Free Dynamic Schedule Recalibration

Traditional schedulers induce "guilt paralysis" by flagging missed tasks in red. PlanZo implements an adaptive rebalancing heuristic:

- **Immovable Time Blocks:** College class schedule (10:30 AM to 5:30 PM) is locked as immutable. The algorithm cannot schedule self-study during lecture slots.
- **Cognitive Weight Priority:** Tasks carry cognitive weights (1 = Light review, 5 = Deep DSA/Math). When evening fatigue is detected, high-weight items are deferred to next morning, and replaced with 25-minute buffer zones.

Algorithm 3: Daily Cognitive Strain & Bandwidth Scoring

COGNITIVE STRAIN INDEX (CSI)

$$\text{CSI} = \text{Sum}(\text{CognitiveWeight}_i * \text{DurationMinutes}_i) / 60$$

If CSI > 6.0: System triggers "Overloaded" status and auto-suggests a 25-minute Chill Buffer.

Algorithm 4: Gamified Day Streak & Non-Linear XP Progression

LEVEL PROGRESSION CURVE

$$\text{Level} = \text{floor}(\sqrt{\text{Total_XP} / 100}) + 1$$

Logarithmic dampening ensures that early engineering milestones feel achievable while high levels require genuine semester-long consistency.

4. Pitch Presentation Guide & Cross-Question Defense

When presenting PlanZo to professors, hackathon panels, or investors, use these battle-tested technical responses to demonstrate engineering maturity.

Q1: Why not just use Google Calendar or Notion for student planning?

Answer: Notion and Google Calendar are blank-canvas productivity tools built for enterprise knowledge workers. They require manual setup and have zero awareness of academic constraints like 75% debarment rules, semester credit distribution, or mid-term PYQ weightage. PlanZo is domain-specific: it calculates attendance debarment margins in real-time, auto-populates B.Tech syllabi, and rebalances routines without guilt.

Q2: How does the application guarantee responsiveness if Gemini API experiences rate-limiting?

Answer: PlanZo employs a multi-tiered resilience strategy. Every AI route in server.ts is wrapped in an asynchronous try/catch block with pre-computed academic heuristic fallback matrices. If the external Gemini API call times out or exhausts quota, the server immediately returns structured, human-like guidance without degrading UI functionality or throwing unhandled 500 errors.

Q3: Why did you build an Express backend rather than a purely static client-side app?

Answer: Security and architectural scalability. Client-side AI integrations leak private API tokens directly into the browser inspection console. By implementing an Express proxy layer, we enforce strict credential isolation, enable request logging, and allow future migration to self-hosted LLMs or multi-tenant database clusters without touching frontend code.

Q4: What mathematical edge cases does the attendance calculator handle?

Answer: The simulator handles boundary cases where $T = 0$ (returning 100% initial margin), fractional rounding via `Math.ceil` and `Math.floor` to ensure students never miss an extra class due to decimal truncation, and supports dynamic target adjustment (e.g. 65% for medical leaves or 80% for honors criteria).

Q5: How does PlanZo safeguard student data privacy?

Answer: All schedule, routine, and habit data is persisted locally in the student browser via the HTML5 LocalStorage API. No personal academic records or habits are sold, tracked, or sent to third-party ad networks.

PLANZO ENGINEERING SYSTEM - VERIFIED FOR PRODUCTION & ACADEMIC DEFENSE